Homework 5: Analaysis of Graphene Hall Bar

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1 Overview

In this exercise, you will analyse the magnetotransport in monolayer graphene. The sample was fabricated by Jacob Amontree (Hone lab) and measured by Dihao Sun (Dean lab) in early 2023. It consists of a sheet of monolayer graphene, shaped into a Hall bar and placed on a SiO_2 substrate. Contacts to measure V_{xx} and V_{xy} simultaneously were added. Applying a gate voltage allows you to control the carrier density in the device, which can be seen as one side of a parallel-plate capacitor.

The first part is the analysis of the Landau fan visible in the $R_{xx}(n, B)$ and $R_{xy}(n, B)$ plots. The second part is the extraction of n and τ_q from the Shubnikov-de Haas oscillations.

The csv (comma separated value) files that accompany this document should work well with any programming language you prefer, but let the TA know if you need the data in another format.

There will be a bonus point if your report is made in LaTeX. Please submit both compilable zip and pdf generated from the tex.

2 Landau fan (all students)

In `fan.csv` you will find six columns of data: the applied magnetic field B (in T), current I (in A), temperature T (in K), gate voltage V_G (in V), longitudinal voltage V_{xx} (in V) and transversal voltage V_{xy} (in V).

- Plot $R_{xx}(V_G, B)$ and $R_{xy}(V_G, B)$. Indicate the filling factor of all quantum Hall states. Explain why we see this sequence.
- Find the conversion between gate voltage V_G and carrier density n. How thick is the dielectric?
- Plot $R_{xx}(\nu, B)$ and explain how the features are arranged. Estimate the sample mobility.
- Plot $\sigma_{xy}(\nu)$ in units of $\frac{e^2}{h}$ at the highest magnetic field. Do you see any hints of symmetry-broken states? Which ones?

3 Shubnikov-de Haas oscillations (PhD students only)

In `sdh.csv` you will find 10 sets of six variables (magnetic field B in T, its inverse in T^{-1}, current I in A, longitudinal voltage V_{xx} in V and longitudinal resistance $R_{xx} = \frac{V_{xx}}{I_{xx}}$ in Ω) at ten different gate biases.

- For every value of gate bias, plot $R_{xx}(1/B)$ and extract the carrier density n. Then, plot $n(V_G)$. How does this compare with the expression from the previous exercise?
- Extract the quantum lifetime τ_q from these oscillations. You can use the definition of momentum $\hbar k_F = m^* v_F$ for the effective mass, where $k_F = \sqrt{\pi n}$ is the Fermi wavevector.
- Estimate the mobility μ from the onset of the oscillations. Plot $\tau(n)$ and $\tau_q(n)$ together and discuss.

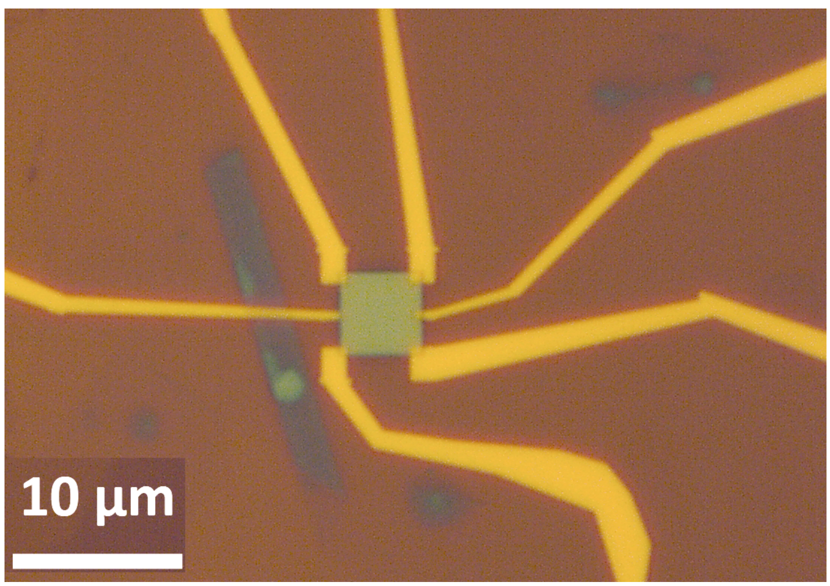

Figure 1: The encapsulated graphene sample the data for this problem set have been taken from.